

Solvency Field Theory V13.11

The Cones and the Ruler

Where the scale of the metric comes from

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Abstract. Causal structure fixes the metric only up to a conformal factor. Null carriers have no rest frame and no intrinsic scale. Massive particles have both. The scale of the metric is therefore not supplied by the light cones, and something else must supply it. This paper identifies what: a persistent closed process carries a natural unit of time, its Compton period, and that unit is the only intrinsic scale any single object has. Sections 2 through 7 stand on established results and require no further commitments. Section 8 proposes one mechanism for why such units exist at all; the argument does not depend on it.

1 The factor nobody supplies

There is a theorem about spacetime that is standard, uncontroversial, and rarely stated in the form that matters here.

The causal structure of a spacetime — the record of which events can influence which — determines the metric *up to a conformal factor*. This is Malament’s result, building on Hawking, King and McCarthy. Know every light cone and you know the geometry’s shape completely: every angle, every cone, every relation of before and after. You do not know a single length.

A conformal structure has no scale. It cannot tell you whether the distance between two events is a millimetre or a megaparsec, because that question has no answer inside it.

So a metric is two things. The cones, and a factor. The cones come from causal structure. The factor comes from somewhere else, and standard physics does not say where.

That is the gap this paper is about.

2 Einstein’s complaint

He noticed it, and he was not happy about it.

In his *Autobiographical Notes* of 1949, Einstein wrote that treating measuring rods and clocks as independent theoretical entities was a sin. His point was that relativity uses them as primitives — the theory tells you how rods contract and clocks dilate, but it does not tell you what a rod or a clock *is*. They should have been derived as solutions of the theory, and they were not.

Harvey Brown made this a full position in *Physical Relativity*. On the dynamical approach, space-time geometry does not explain the behaviour of matter; the behaviour of matter explains the geometry. Rods contract because of what they are made of and how it holds together, not because space instructs them to.

Both objections are about the same missing piece. A rod is a length standard. A clock is a duration standard. A standard is a scale. And the scale is exactly what causal structure does not provide.

3 What a null carrier cannot supply

Take the gap seriously and ask the obvious question: could light supply the scale?

It cannot, and the reason is geometric rather than practical.

A null worldline has no rest frame. No Lorentz transformation takes a null four-vector to zero spatial momentum, so there is no frame in which a photon sits still and everything else is measured against it. This is not a limitation of technique. It is a fact about the structure of the Lorentz group.

Wigner’s 1939 classification makes the consequence exact. The little group — the subgroup of Lorentz transformations that leaves a particle’s momentum alone — differs between the two cases:

carrier	little group	spin states
massive	$SO(3)$, the full rotation group	$2s + 1$
massless	$ISO(2)$, the Euclidean plane group	2, helicity $\pm s$

A massive particle has a rest orientation. A photon does not, and the difference is measured: a photon has two polarisation states, not three. A massive vector particle has three. The missing state is the missing rest frame.

A photon’s only Poincaré invariant is $p \cdot p = 0$. Its energy and momentum are frame-dependent, so they are not properties it carries; they are relations to whoever is looking. Strip away every observer and a photon retains its helicity and nothing else.

Null carriers give the cones. They cannot give the ruler.

4 What a closed process supplies

A massive particle supplies three things that light cannot.

A rest frame. There is a frame in which it sits still. That frame is a “here” — a place from which other things can be located.

A tally. A closed process returns to itself, and each return is a countable event. That count is proper time. A photon has no return, so it has no count; its elapsed time along any null path is exactly zero, not merely small.

A unit. This is the one that matters for the gap. A massive particle has a natural period, and only one:

$$T_C = \frac{h}{mc^2}, \quad \ell_C = cT_C = \frac{h}{mc}.$$

This is the Compton period, and it is not a convention. It is fixed by the particle’s mass, it is the same for every particle of that species anywhere in the universe, and it is the only intrinsic time any single object possesses.

	unit of time	unit of length
electron	8.0933×10^{-21} s	2.4263×10^{-12} m
muon	3.9142×10^{-23} s	1.1734×10^{-14} m
proton	4.4077×10^{-24} s	1.3214×10^{-15} m

Those are rulers. Not analogies for rulers — lengths, with values, carried by every particle of the species, requiring no calibration and no agreement with anyone.

5 The ladder

Put the pieces in order and the dependency structure is stark.

Level 0 — null carriers only. Present: light cones, causal order, conformal geometry, helicity. Absent: any rest frame, any tally, any unit, any “here”, the metric scale. The geometry is fixed in shape and undetermined in size.

Level 1 — one closed process. It adds a rest frame, a tally, a unit, and the Casimir invariants: mass, charge, spin magnitude. Combined with level 0, conformal structure plus a scale is a metric. One object completes it, locally.

Still absent at this level: position, velocity, spin direction, separation, gravitational interaction. All of those need something to be relative to.

Level 2 — two closed processes. Separation becomes definable. So do relative velocity, relative orientation, and therefore spin direction. Mutual gravitation appears, because gravitation is an interaction and interactions need two. And tallies can be compared, so time dilation becomes observable rather than merely true.

Measurement exists at level 2, because measurement is comparison.

Level 3 — many. Rigid extension, since a rod is many closures at fixed separation. Superposed gravitation. Thermodynamics. Horizons.

6 What this means

The apparatus physics writes its equations in — coordinates, frames, rods, clocks, metrics — is not a stage that matter was later placed upon. It is assembled, level by level, out of what persists.

Causal structure gives shape without size. A closed process gives size. Two closed processes give relation. Many give extension.

Physics writes its equations in a language that exists because of the things the equations describe.

7 Where the scale can come from

One dimensionful input is needed and there are exactly two known candidates.

Matter. Any massive species carries h/mc^2 , as above.

The cosmological constant. Λ has dimensions of inverse length squared and supplies the de Sitter radius $\sqrt{3/\Lambda} = 1.65 \times 10^{26}$ m, about 17 billion light years.

Both are dimensionful, and neither comes from causal structure. This paper does not claim matter is the only possible source. It claims that the source is not the cones, that the two known candidates are matter and Λ , and that in any region containing matter the local scale is set by matter, because the Compton period is local and $\sqrt{3/\Lambda}$ is not.

Whether those two are independent or related is open, and it is a better question than it looks.

8 One mechanism for the unit

Everything above stands on established results. This section does not, and the argument does not require it.

The natural question left by §4 is why a massive particle has an intrinsic period at all. Standard physics gives the relation $T_C = h/mc^2$ and does not say what is happening.

Solvency Field Theory proposes an answer: a massive particle is a null carrier held in a closed route. The carrier moves at c , as null carriers do; the route returns it to itself; and one traversal is one completed event. The period is then not a postulate but a consequence of the route's length, and the transition from level 0 to level 1 becomes a physical process rather than a definition — something null becomes something closed, and a unit appears where there was none.

On that reading the ladder is not a taxonomy. It is a history. The universe did not begin with rulers.

This is one candidate. Others may exist, and the levels above would be unchanged by a different mechanism supplying the same unit. What the argument requires is that *something* supplies it; what SFT offers is a specific proposal for what.

9 Limits

The scale is local. One closed process fixes a unit where it is. Extending that to a metric everywhere requires transporting the unit, which needs either many processes or an assumption of homogeneity. The level-1 claim is local and is stated that way.

Vacuum solutions remain valid. A matter-free solution of the Einstein equations is a perfectly good mathematical object with a perfectly good metric. Nothing here says matter is necessary for a metric to exist. The claim concerns what fixes the scale of a metric that describes a physical situation, and in vacuum regions of a universe containing matter, the scale is inherited.

The dimensional argument does not establish uniqueness. Showing that causal structure cannot supply a scale, and that two known things can, is not a proof that only those two can.

And §8 is a proposal. It is marked as one, it is short for that reason, and removing it leaves §1–§7 intact.

10 Claim boundary

This paper does not derive general relativity, quantum mechanics, or any constant. It does not claim the metric requires matter to exist. It does not claim to be the first relational account of measurement; Einstein raised the problem, Brown developed it, and this is a contribution to that line rather than a departure from it.

It claims three things, and all three are composed of established results:

Causal structure fixes the metric only up to a conformal factor. Null carriers cannot supply the

missing factor, and the two-state polarisation of the photon is the observable signature of why not. A closed massive process supplies it, and the value of the unit it supplies is h/mc^2 .

What follows from those three — that the measuring apparatus of physics is constituted rather than presupposed — is stated once, in §6, and is left to the reader from there.

Status. V13.11. Sections 1–7 and 9–10 stand independently of Solvency Field Theory. Section 8 is a mechanism proposal and is separable.

Originator and lead author: Shaun Fosmark. Narrative and argument construction: Claude/Opus. Derivation and audit canon: GPT-5.6 SOL.